

Computer Networks I

Error Detection

(Parity check and Internet checksum)

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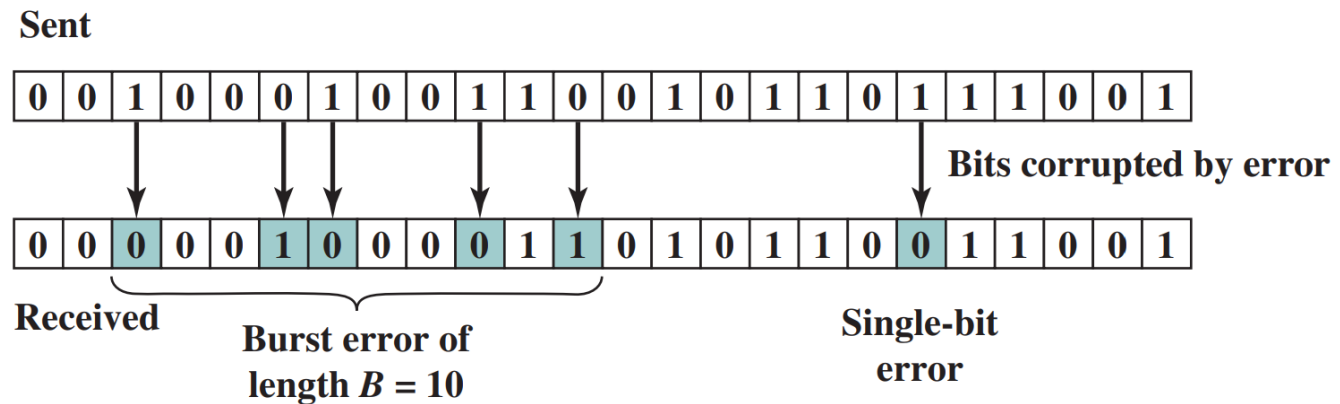
Types of Errors

- An error occurs when a bit is altered between transmission and reception
 - Types of errors:
 - Single bit errors
 - Burst errors
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Types of Errors

- **Single bit error:**

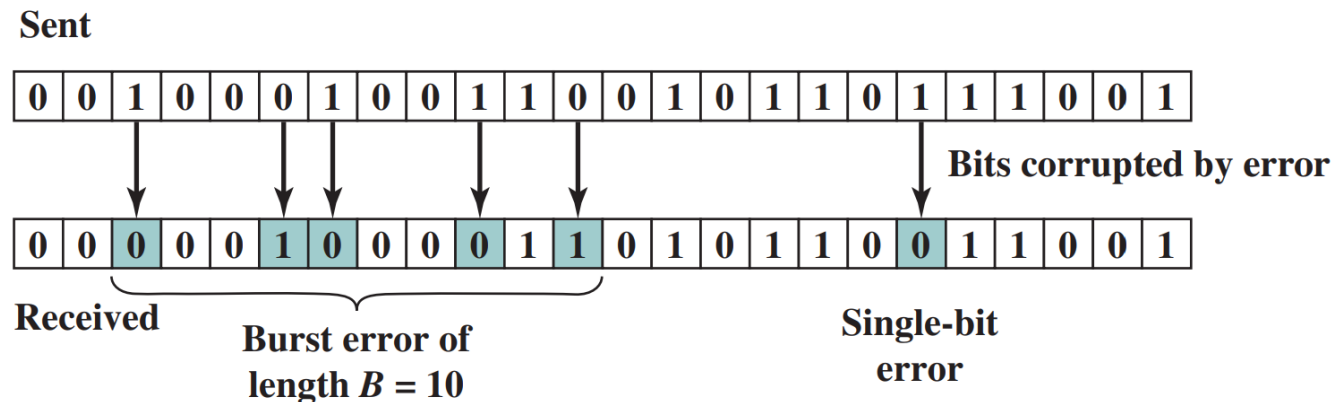
- Isolated error that alters one bit but does not affect nearby bits
- Can occur in presence of white noise



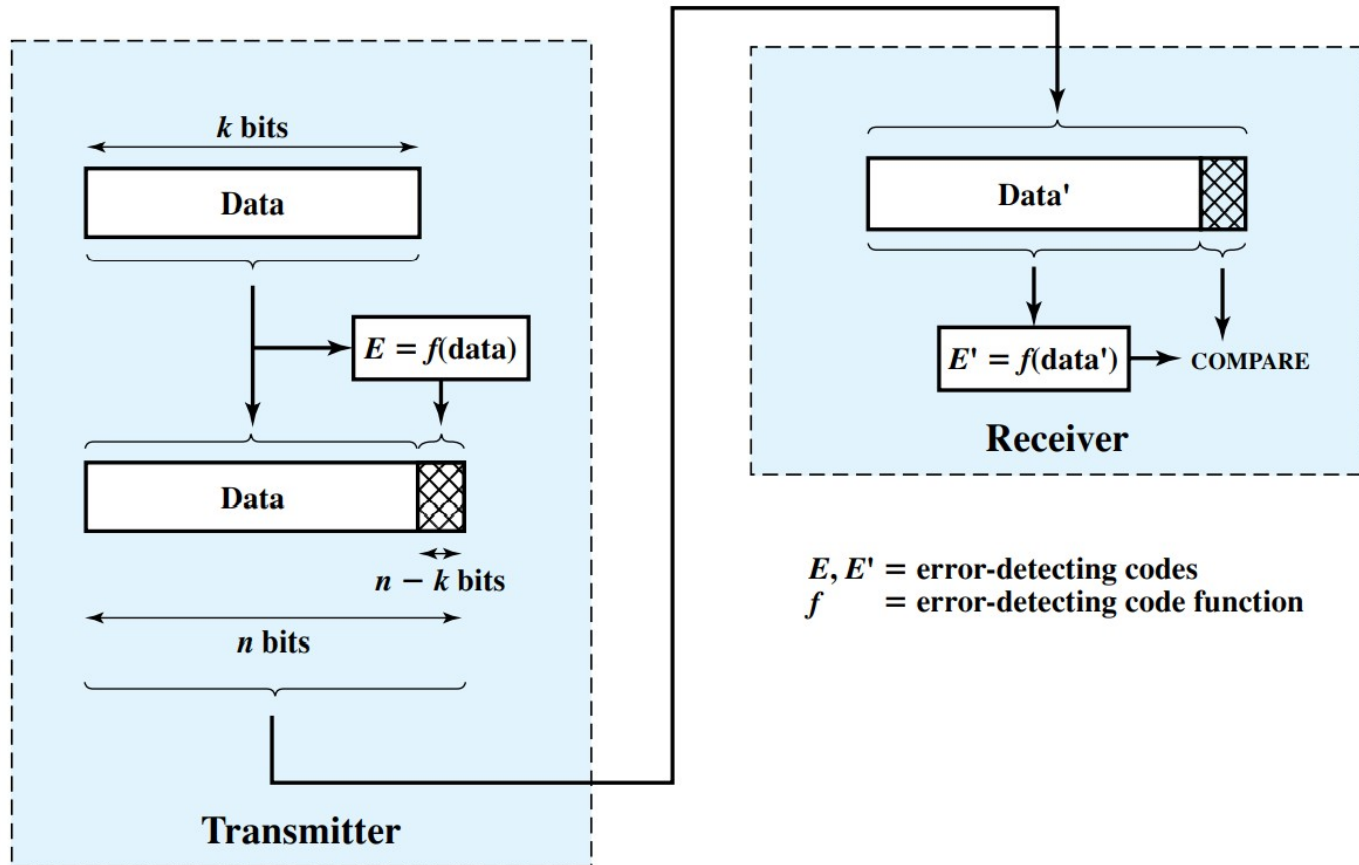
Types of Errors

- **Burst error:**

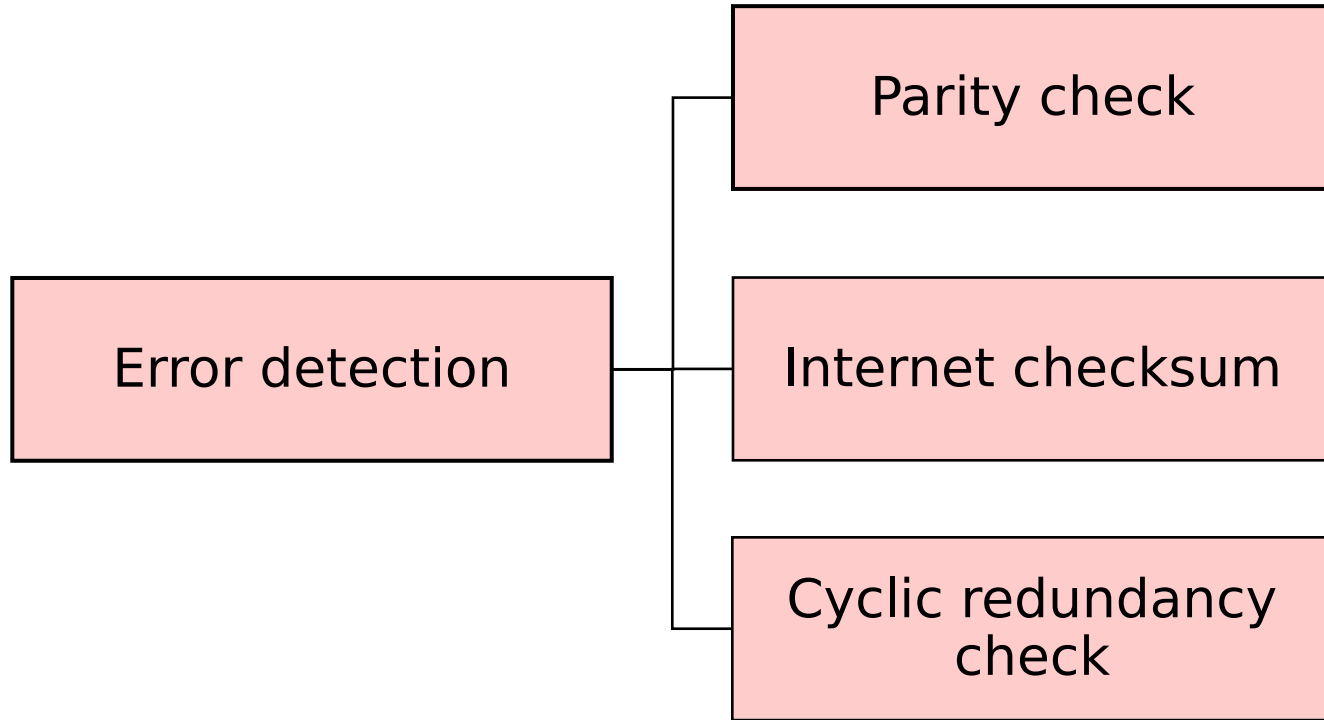
- Contiguous sequence of B bits in which the first and last bits and any number of intermediate bits are received in error
- Can occur due to impulse noise or fading in wireless environment
- Effects are greater at higher data rates



Error Detection



Error Detection



Parity Check

- The simplest error detecting scheme is to append a parity bit to the end of a block of data
 - Even parity: Total number of ones will be even
 - Odd parity: Total number of ones will be odd
 - Any odd number of bit errors is detected
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Parity Check

- ❑ **Limitation:** If any even number of bits are inverted due to error, an undetected error occurs
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Two-dimensional Parity Check

- 01110 01110 01000 01011

			Row parity →					
	$b_{1,1}$	• • •	$b_{1,j}$	r_1				
	$b_{2,1}$	• • •	$b_{2,j}$	r_2				
	$b_{i,1}$	• • •	$b_{i,j}$	r_i				
Column parity ↓								
	c_1	• • •	c_j	p				

0	1	1	1	0	1
0	1	1	1	0	1
0	1	0	0	0	1
0	1	0	1	1	1
0	0	0	1	1	0

- 01110**1** 01110**1** 01000**1** 01011**1** 000110

Two-dimensional Parity Check

- A single-bit error is detected and corrected
- Any odd number of bit errors is detected
- Some even number of bit errors is also detected

0	1	1	1	0	1
0	0	1	1	0	1
0	1	0	0	0	1
0	1	0	1	1	1
0	0	0	1	1	0

Two-dimensional Parity Check

- Any pattern of four errors forming a rectangle is undetectable

0	1	1	1	1	1	0	1
0	0	1	1	0	1	1	0
0	0	1	1	0	0	1	1
0	0	0	0	0	0	0	0
1	0	1	1	1	1	1	0
1	1	0	0	0	1	1	0

The Internet Checksum

- Ones-complement operation:
 - Replace 0 with 1, and 1 with 0
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The Internet Checksum

- Ones-complement addition:
 - The two numbers are treated as unsigned binary integers and added
 - If there is a carry out of the leftmost bit, add 1 to the sum (end-around carry)
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The Internet Checksum

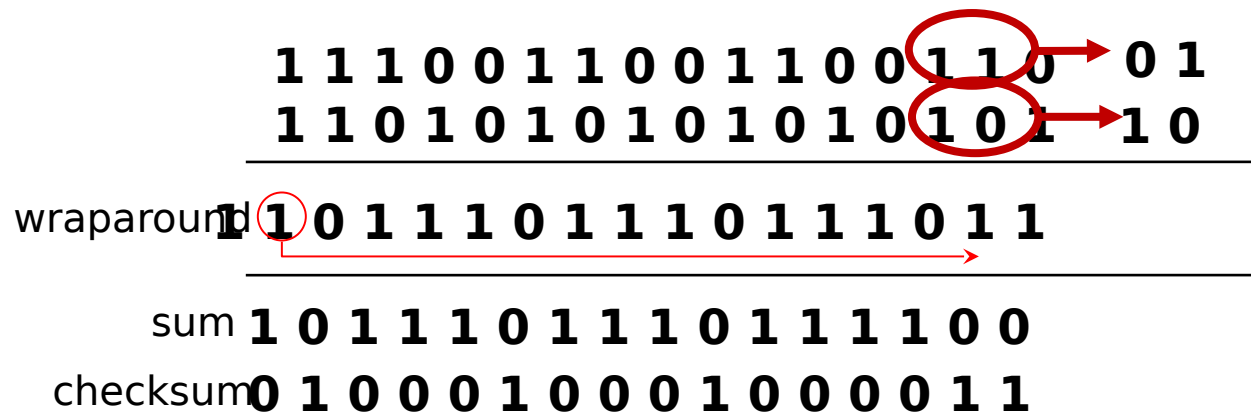
- 00 01 F2 03 F4 F5 F6 F7 00 00

Partial sum	$\begin{array}{r} 0001 \\ F203 \\ \hline F204 \end{array}$
Partial sum	$\begin{array}{r} F204 \\ F4F5 \\ \hline 1E6F9 \end{array}$
Carry	$\begin{array}{r} E6F9 \\ 1 \\ \hline E6FA \end{array}$
Partial sum	$\begin{array}{r} E6FA \\ F6F7 \\ \hline 1DDF1 \end{array}$
Carry	$\begin{array}{r} DDF1 \\ 1 \\ \hline DDF2 \end{array}$
Ones complement of the result	220D

Partial sum	$\begin{array}{r} 0001 \\ F203 \\ \hline F204 \end{array}$
Partial sum	$\begin{array}{r} F204 \\ F4F5 \\ \hline 1E6F9 \end{array}$
Carry	$\begin{array}{r} E6F9 \\ 1 \\ \hline E6FA \end{array}$
Partial sum	$\begin{array}{r} E6FA \\ F6F7 \\ \hline 1DDF1 \end{array}$
Carry	$\begin{array}{r} DDF1 \\ 1 \\ \hline DDF2 \end{array}$
Partial sum	$\begin{array}{r} DDF2 \\ 220D \\ \hline FFFF \end{array}$

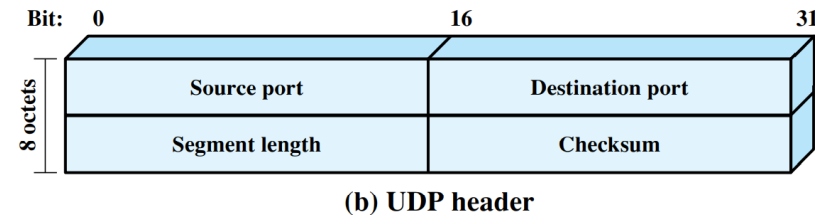
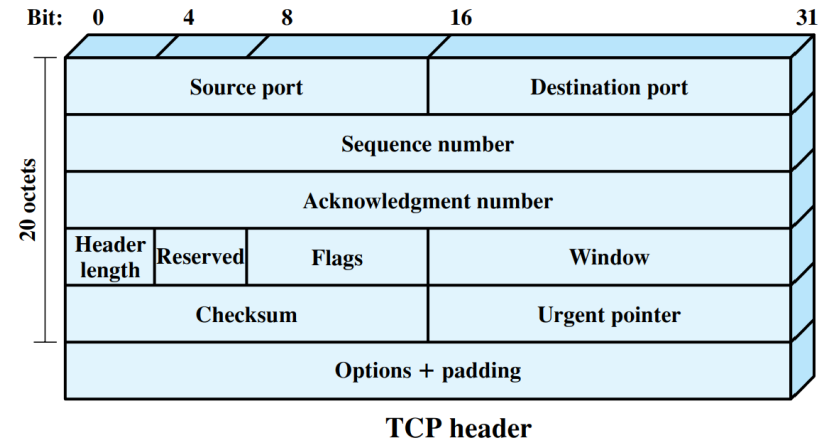
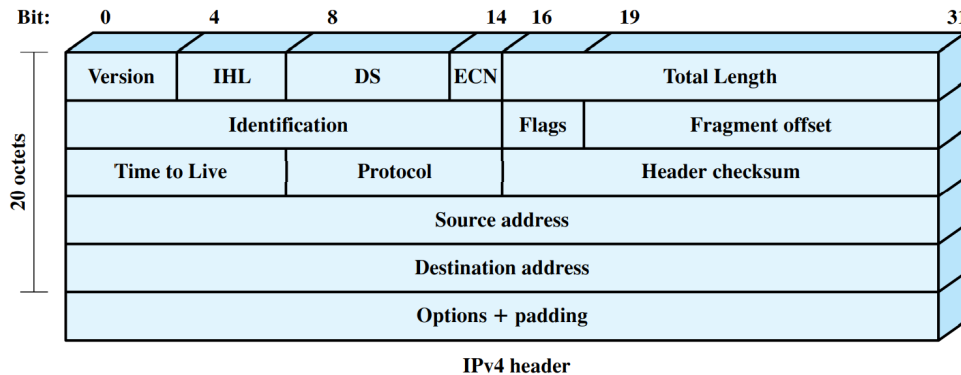
The Internet Checksum

- Provides better greater error detection capability than parity check
 - Provides **weak** protection



The Internet Checksum

- Internet checksum is used in many Internet standard protocols, including IP, TCP, and UDP



Summary

□ Different error detection techniques discussed:

- Parity check
 - Internet checksum
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